CS & IT ENGINEERING

Digital Logic

Boolean Theorems and Logical Gates

Lecture No. 5

By- Aditya sir

Recap of Previous Lecture

Topic

XOR

Properties

Practice

Topics to be Covered

Topic

XNOR

* Properties of XOR & XNOR.

About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.

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XOR

$$A \oplus B = \bar{A}B + A\bar{B}$$

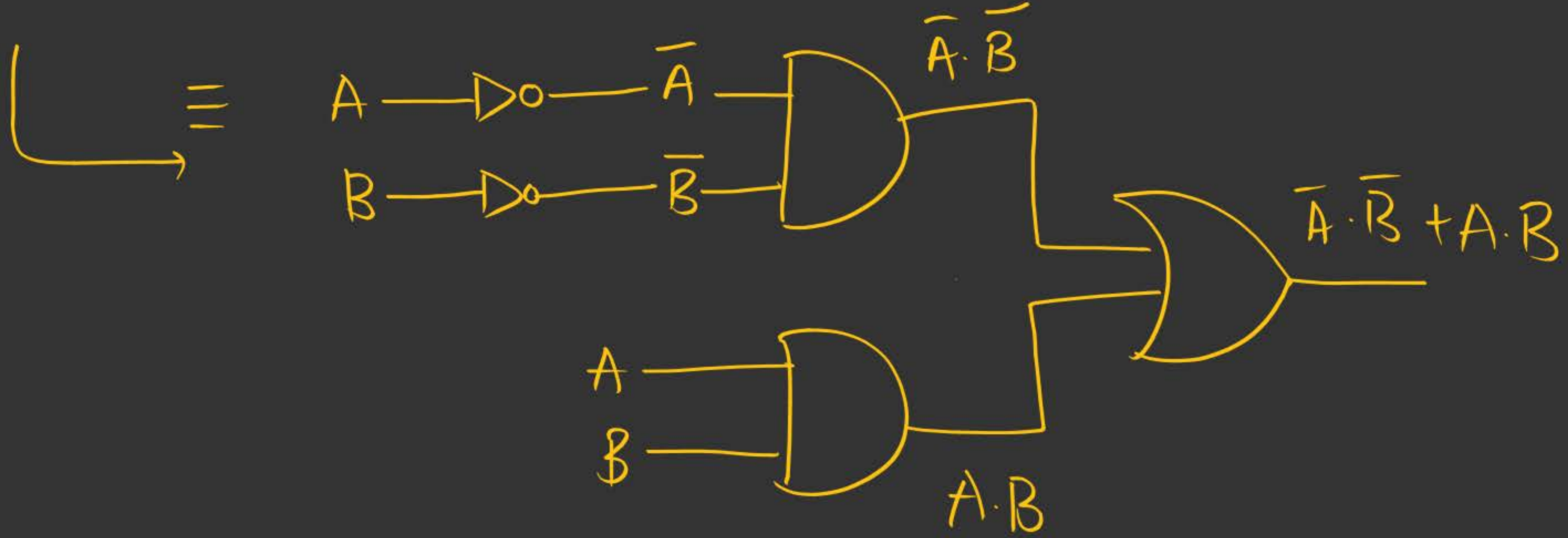
A	B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

XNOR gate : Exclusive NOR

Repr :-

$$A \odot B = \bar{A} \cdot \bar{B} + A \cdot B$$

$$A \odot B = \bar{A}\cdot\bar{B} + A\cdot B$$



Truth table

	A	B	$A \odot B = \bar{A}\bar{B} + A \cdot B$
0	0	0	1
1	0	1	0
2	1	0	0
3	1	1	1

$$0\ 0 = \bar{0} \cdot \bar{0} + 0 \cdot 0 = 1 \cdot 1 + 0 = 1$$

$$0\ 1 = \bar{0} \cdot \bar{1} + 0 \cdot 1 = 1 \cdot 0 + 0 \cdot 1 = 0$$

$$1\ 0 = \bar{1} \cdot \bar{0} + 1 \cdot 0 = 0 + 0 = 0$$

$$1\ 1 = \bar{1} \cdot \bar{1} + 1 \cdot 1 = 0 + 1 = 1$$

$A = 1$
$\bar{A} = 0$

SOP

$$\Sigma(0, 3)$$

0 0   1 1

$$(\bar{A} \cdot \bar{B} + A \cdot B)$$

$A = 0$
$\bar{A} = 1$

POS $\Rightarrow \Pi(1, 2)$

0 1   1 0

$$(A + \bar{B}) \cdot (\bar{A} + B)$$

$$\overline{A \odot B} = \overline{(\bar{A}+B)(A+\bar{B})}$$

$$\boxed{\overline{A \oplus B} = A \odot B}$$

$$= \overline{(\bar{A}+B)} + \overline{(A+\bar{B})}$$

$$= A\cdot\bar{B} + \bar{A}\cdot B = A \oplus B$$

$$\overline{p \odot q} = p \oplus q$$

$$p \odot q = \bar{p} \cdot \bar{q} + p \cdot q$$

$$\bar{p} \odot q = \bar{\bar{p}} \cdot \bar{q} + \bar{p} \cdot q = p \cdot \bar{q} + \bar{p} \cdot q = p \oplus q$$

$$p \odot \bar{q} = \bar{p} \cdot \bar{\bar{q}} + p \cdot \bar{q} = \bar{p} \cdot q + p \cdot \bar{q} = p \oplus q$$

$$\bar{p} \odot \bar{q} = \bar{\bar{p}} \cdot \bar{\bar{q}} + \bar{p} \cdot \bar{q} = p \cdot q + \bar{p} \bar{q} = p \odot q$$

① Commutative Law:

$$\boxed{A \odot B = B \odot A}$$ → Holds Commutative law

② Association law

Holds ✓

$$A \odot B \odot C = (A \odot B) \odot C = A \odot (B \odot C)$$

* Properties of XNOR

1) $A \odot A = 1$

2) $\bar{A} \odot \bar{A} = 1$

3) $A \odot \bar{A} = \bar{A} \odot A = 0$

$$A \odot A = 1 \Rightarrow A \odot 1 = A$$

Proof:- $A \odot 1 \Rightarrow \bar{A} \cdot \bar{1} + A \cdot 1 = 0 + A = A$

$$A \odot \bar{A} = 0 \Longrightarrow A \odot 0 = \bar{A} \Rightarrow$$

Proof:- $A \odot 0 = \bar{A} \cdot \bar{0} + A \cdot 0$

$= \bar{A} \cdot 1 + 0 = \bar{A}$

$P Q \odot 0 = \overline{PQ}$

$= \bar{P} + \bar{Q}$

Exchange Property :- in general XNOR.

if $A \odot B = C \implies \underline{\underline{A \odot C}} = B$

Proof :- given : $A \odot B = C$

$$A \odot C = A \odot (A \odot B) = (A \odot A) \odot B$$

$$= 1 \odot B$$

$$= B$$

$$A \odot B = C \Rightarrow B \odot A = C \Rightarrow \underline{B \odot C = A}$$

$A \odot A \odot A \odot \cdots\cdots$ n times

the o/p is ________

if $\underline{n = \text{even}} \Rightarrow \quad = 1$

$n = \text{odd} \rightarrow \quad = A$

$1 \odot A = A$

$$A \odot A = 1$$

$$(A \odot A) \odot A = 1 \odot A = A$$

$$(A \odot A) \odot (A \odot A) = 1 \odot 1 = 1$$

Buffer & inverted Buffer.

$$\left.\begin{aligned} A \odot 0 &= \bar{A} \\ A \odot 1 &= A \end{aligned}\right\} \underline{\text{enable}}$$

control $= 0 \rightarrow A \odot 0 = \bar{A}$

Control $= 1 \Rightarrow A \odot 1 = A$

Control $= 0 \longrightarrow$ inverted Buffer

$= 1 \longrightarrow$ Buffer

Combined properties of XOR & XNOR

1) $\overline{A \oplus B} = A \odot B$

2) $\bar{A} \oplus B = \bar{\bar{A}} \cdot B + \bar{A}\,\bar{B} = A \cdot B + \bar{A}\,\bar{B} = A \odot B$

3) $A \oplus \bar{B} = \bar{A} \cdot \bar{B} + A \cdot \bar{\bar{B}} = A \cdot B + \bar{A} \cdot \bar{B} = A \odot B$

☆ 4) $\bar{A} \oplus \bar{B} = \bar{\bar{A}} \cdot \bar{B} + \bar{A}\,\bar{\bar{B}} = A \cdot \bar{B} + \bar{A} \cdot B = A \oplus B$

$$\boxed{\overline{A \oplus B} = \bar{A} \oplus B = A \oplus \bar{B} = A \odot B}$$

$$\text{1) } \overline{A \odot B} = A \oplus B$$

$$\text{2) } \bar{A} \odot B = \bar{\bar{A}} \cdot \bar{B} + \bar{A} \cdot B = A\bar{B} + \bar{A}B = A \oplus B$$

$$\text{3) } A \odot \bar{B} = \bar{A} \cdot \bar{\bar{B}} + A \cdot \bar{B} = \bar{A} \cdot B + A\bar{B} = A \oplus B$$

$$\text{4) } \bar{A} \odot \bar{B} = \bar{\bar{A}} \cdot \bar{\bar{B}} + \bar{A} \cdot \bar{B} = \bar{A} \cdot \bar{B} + A \cdot B = A \odot B$$

$$\boxed{\overline{A \odot B} = \bar{A} \odot B = A \odot \bar{B} = A \oplus B}$$

$$\bar{A} \odot \bar{B} = A \odot B$$

(Q) $A \oplus B \oplus C = \overline{A \odot B \odot C}$? T/F → **False**

A	B	C	$A \oplus B \oplus C$	$A \odot B \odot C$	
0	0	0	0	0	0
0	0	1	1	1	1
0	1	0	1	1	2
0	1	1	0	0	3
1	0	0	1	1	4
1	0	1	0	0	5
1	1	0	0	0	6
1	1	1	1	1	7

$$f(A,B,C) = \Sigma(1,2,4,7)$$

1) $A \oplus B \oplus C = A \odot B \odot C$

2) $A \oplus B = \overline{A \odot B}$

2 $1 \oplus 1 = 0$

$\boxed{1}$ $1 \oplus 0 = 1$

2 $1 \oplus 0 \oplus 1 = 0$

2 $1 \oplus 1 \oplus 0 = 0$

$\boxed{3}$ $1 \oplus 0 \oplus 0 \oplus 1 \oplus 1 = 1$

0 $0 \oplus 0 \oplus 0 = 0$

obs:- for any no. of i/p's
if no. of 1's are odd
then XOR will give 1,
else 0.
→ no. of 1's = even to 0

5 1's
→ odd 1's = 1

$1 \oplus 1 \oplus 0 \oplus 1 \oplus 1 \oplus 0 \oplus 0 \oplus 0 \oplus 1 = \boxed{1}$

XNOR

	no. of i/ps	no. of 1's
1) 1 0 1 = 1	2	2
2) 1 0 1 0 1 = 1	3	3
3) 1 0 1 0 0 = 0	3	2
4) 0 0 1 0 1 0 1 = 0	4	3
5) 1 0 0 0 1 0 0 = 1	4	2
6) 1 0 1 0 0 0 1 0 1 = 0	5	4
7) 1 0 1 0 0 0 1 0 0 = 1	5	3
8) 0 0 0 = 1	2	0
9) 0 0 0 0 0 = 0	3	0

XNOR

same i/p → 1

diff i/p → 0

Imp : XNOR is 1 if

Case 1 :- Even no. of i/p's & no. of 1's are also even

Case 2 : Odd no. of i/ps & no. of 1's are also odd.

AJ Sir's Matrix Pattern

Conclusion

XOR

i/ps \ 1's	E	Od
E	0	1
Od	0	1

XNOR

i/ps \ 1's	E	od
E	1	0
od	0	1

★ Imp

→ for odd no. of i/p's → both XOR and XNOR are same.

→ For even no. of i/p's → XOR & XNOR are Complement of each other.

1) $\overline{A \oplus B} = A \odot B$

2) $A \oplus B \oplus C = A \odot B \odot C$

3) $\overline{A \oplus B \oplus C \oplus D} = A \odot B \odot C \odot D$

(Q)

$\bar{A} \oplus B = A \odot B$

then $y = ?$

A) Always 0

B) $A \oplus B$

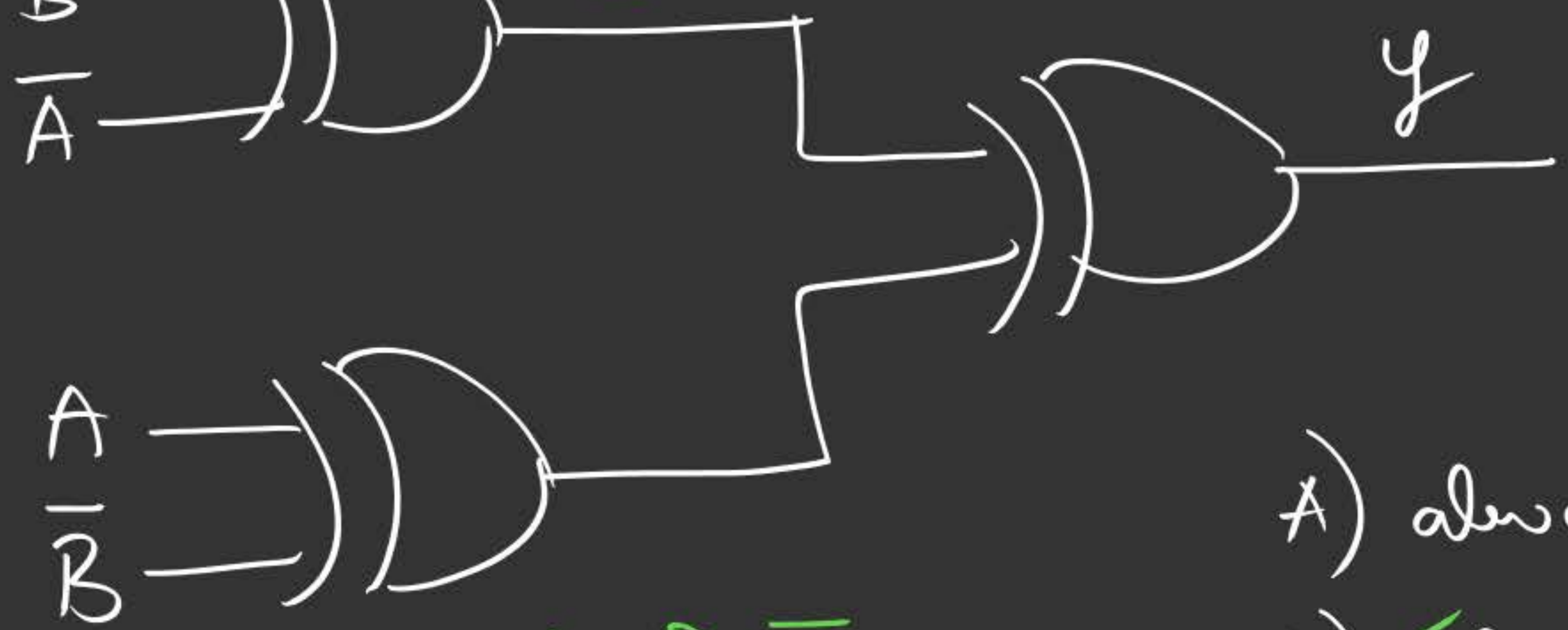
C) $A \odot B$

D) Always 1.

$\bar{B} \oplus A = A \odot B$

Let $A \odot B = C$

$y = (A \odot B) \oplus (A \odot B) \Rightarrow C \oplus C = 0$



$\overline{B} \oplus \overline{A} = A \oplus B$

$A \oplus \overline{B}$

$= A \odot B$

A) always 0

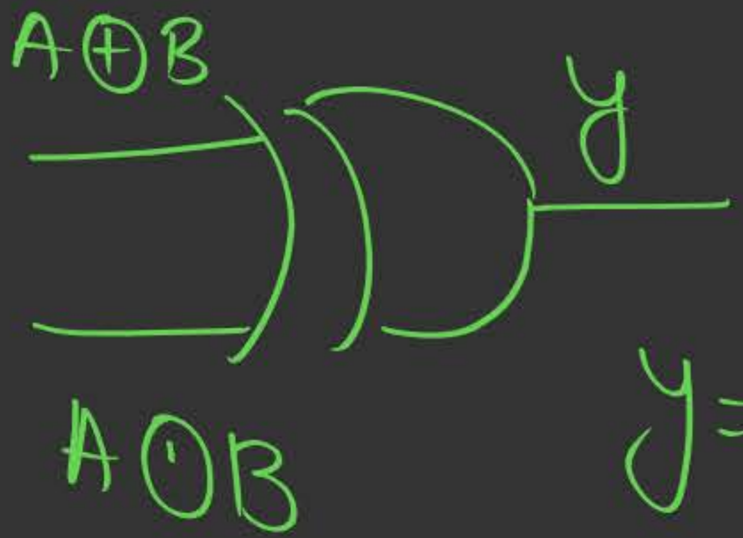
B) always 1

C) $A \oplus B$

D) $A \odot B$

let $A \oplus B = C$

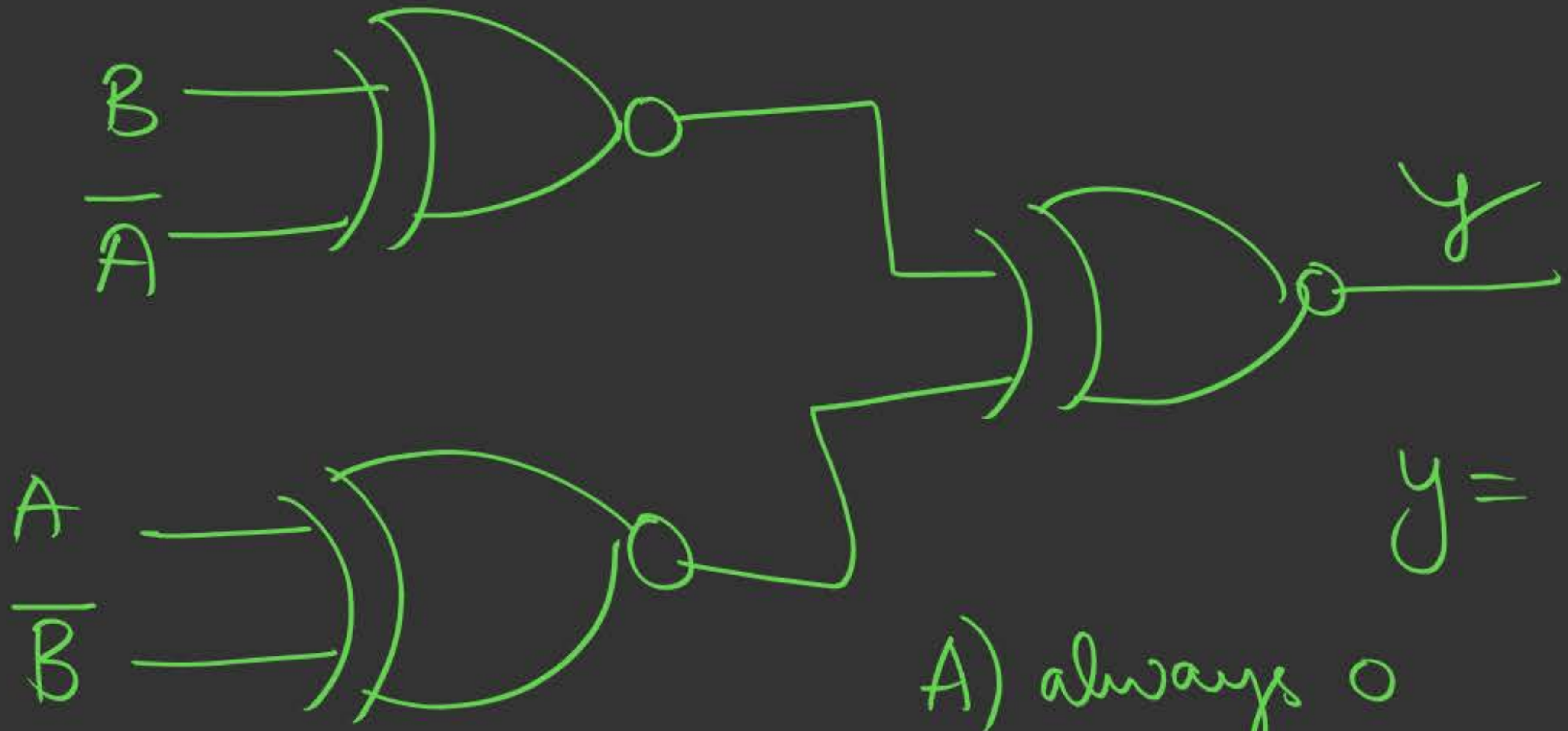
$A \odot B = \overline{C}$



$y = (A \oplus B) \oplus (A \odot B)$

$= C \oplus \overline{C} = 1$

HW 1



$y = ?$

A) always 0

B) $A \oplus B$

C) $A \odot B$

D) always 1

HW2

$$y = \overline{\overline{A \oplus B} \oplus \overline{C}} \quad \text{then } y = ?$$

A) $A \odot B \oplus C$

B) $\overline{\overline{A} \oplus \overline{B \odot C}}$

C) $\overline{A \oplus B \oplus \overline{C}}$

D) $A \oplus B \oplus C$

2 mins Summary

Topic XOR

Topic XNOR

Topic Properties of XOR & XNOR

Topic Practice Ques

Topic

THANK - YOU